

CHEM 427/527 · SESSION 2 · TUESDAY, OCTOBER 6, 2026

Python with chemical data

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Drafted with assistance by Claude Code. Reviewed and revised by Matt.

LEARNING OBJECTIVES

By the end of today you can

1. Index a list of measurements, and select values from it with a loop and with a list comprehension.
2. Write a function whose docstring states the unit of every argument and of the result.
3. Compute the residuals of a calibration fit, and use a residual plot to decide which standards belong in the fit.
4. Use Claude Code to explain an error message, and name the line it points to before you change any code.

FIRST, YOUR NOTEBOOK

Everyone runs their own notebook

On your laptop

Complete and submit your own **activity.ipynb**.

Talk it through with your neighbors as you go.

Laptop trouble?

Tell your TA: Lucas (A to K) or Chris (L to Z).

Until it is fixed, pair with a neighbor: one notebook, both names.

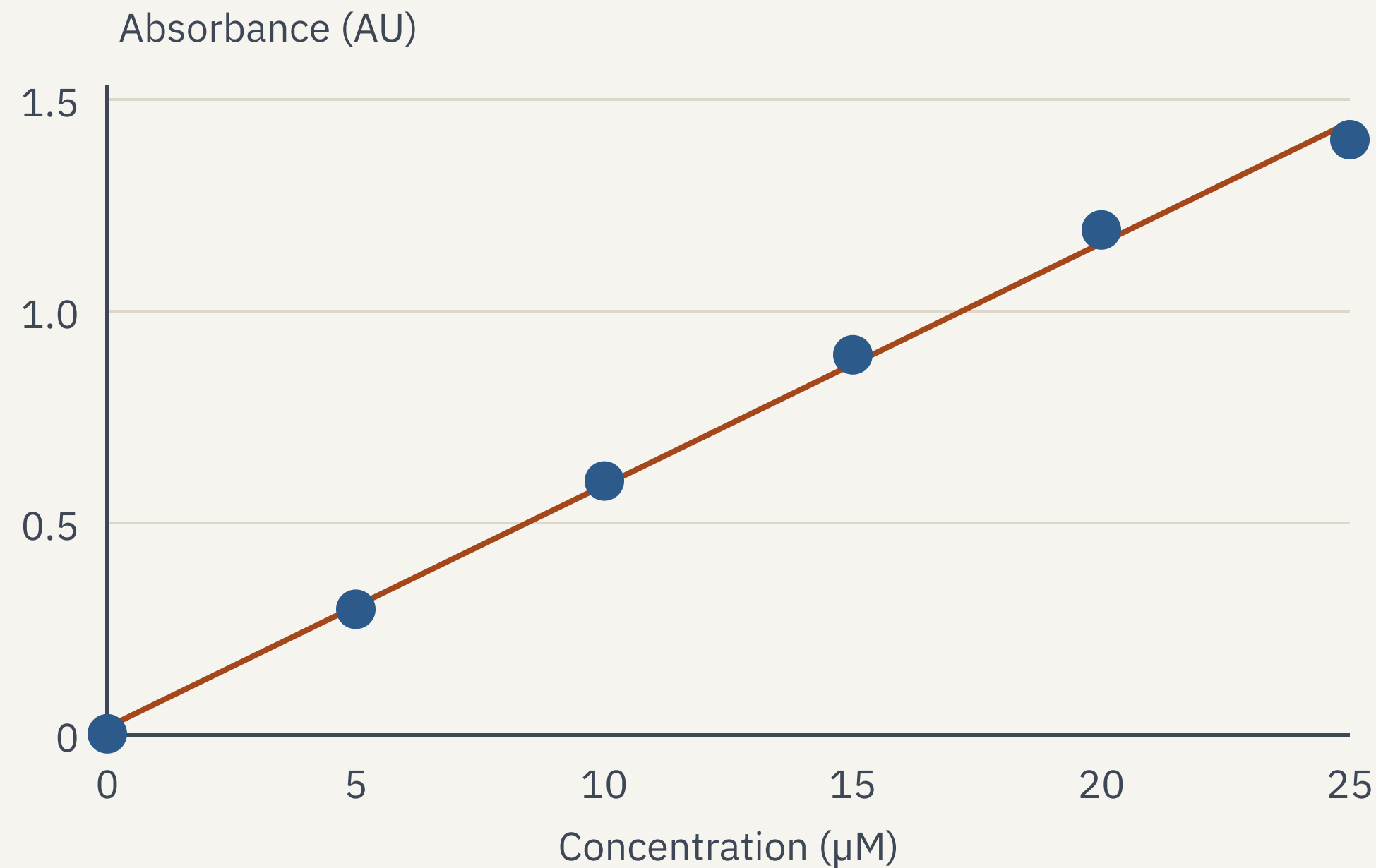
Right now

Run **git pull** in the course folder.

A new folder, **S02-python-with-chemical-data**, appears.

WHERE SESSION 1 STOPPED

Six standards and a straight line



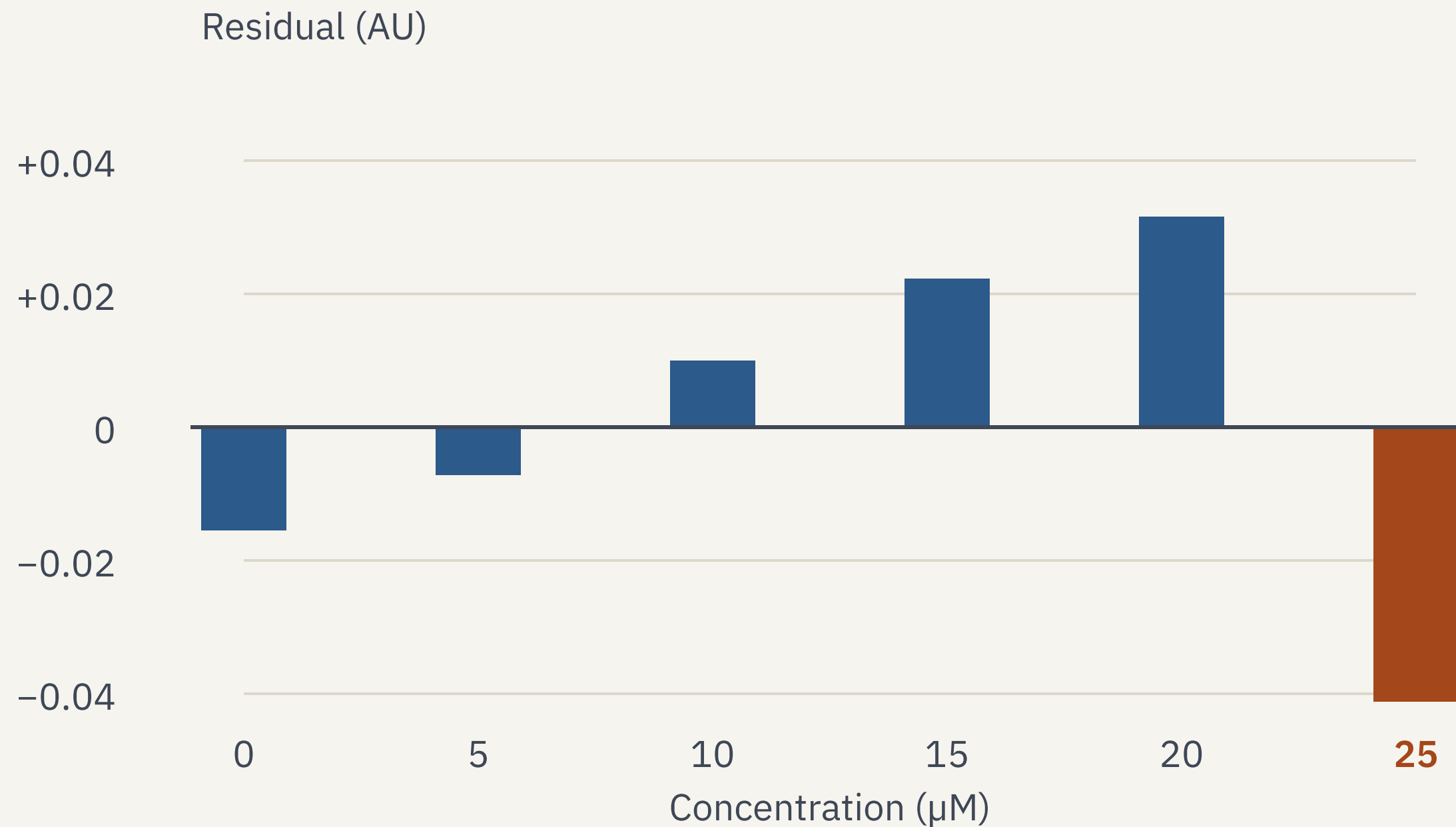
Slope **0.0571 AU/μM**

$R^2 = 0.9975$

Is this a good fit?

WHERE SESSION 1 STOPPED

The residuals curve



Residual: measured absorbance minus the absorbance the line predicts.

They rise from -0.016 to $+0.032$ AU, then drop to -0.041 AU at $25\text{ }\mu\text{M}$.

A curve in the residuals means the model is wrong.

Four pieces of Python

1. Lists, counted from 0

```
absorbances[5]
```

The sixth of the six standards.

2. Loops and comprehensions

```
[a for a in absorbances if a <= 1.2]
```

Caffeine peak list: 5 peaks at a 5% threshold, 12 at 1%.

3. Functions with docstrings

```
def arrhenius(...):
```

Rate constants from 283 to 343 K, with units in the docstring.

4. Arrays, a mask, linregress

```
absorbance <= LINEAR_LIMIT
```

The residual plot from the last slide, which you make in section 4.

FUNCTIONS WITH DOCSTRINGS

The unit mistake prints nothing

Activation energy in J/mol, as the docstring says

80.0e3

8.91 × 10⁻³ 1/s

Measured at 313 K: 9.05 × 10⁻³ 1/s

The same energy typed in kJ/mol

80.0

1.94 × 10¹¹ 1/s

2 × 10¹³ times too large. No error.

An error message is the easy case.

Read the traceback from the bottom

1. The last line

The type and the message:
IndexError: list index out of range

2. The arrow

The line that failed. Name it before you change any code.

3. Ask to explain

Paste the whole message into Claude Code with the prompt below. You write the fix.

The prompt, from the notebook

```
Explain this error message from section 1 of sessions/S02-python-with-chemical-  
data/activity.ipynb. Say which line it points to and why it fails. Do not fix it.
```


BEFORE THURSDAY

Submit a notebook that runs

Submit

Restart, Run All, save.

Upload **activity.ipynb**
by **1:30 pm Thursday,**
October 8.

Scoring

3 if it runs top to bottom
after a restart, **0** if not.

Explain cells are not
graded.

Third hour

Setup clinic with Lucas
and Chris.

Working install by
Tuesday, October 13.